

End-to-End Analysis Case Study

Financial Sample Dataset — Sales, Discounts & Profitability

This report documents an end-to-end analysis pipeline: data cleaning in Python, exploratory analysis using SQL and Pandas, visualization with Matplotlib/Seaborn, and business recommendations derived from a 700-record financial sales dataset covering five countries, five customer segments, and six products.

Dataset	Financial Sample (Microsoft) — 700 orders, 16 fields
Scope	5 Countries · 5 Segments · 6 Products · Sep 2013 – Dec 2014
Tools	Python (Pandas), SQLite (SQL), Matplotlib/Seaborn
Deliverables	Cleaned dataset, SQL/Pandas analysis, charts, this report

1. Data Cleaning & Validation

The raw dataset contained 700 rows and 16 columns with no duplicate records. The only missing values were 53 blanks in **Discount Band**; investigation confirmed every one of these rows had a Discounts value of exactly 0, so they were relabeled as '**None**' rather than dropped or imputed.

All three calculated fields were cross-checked against their formulas and found to be 100% internally consistent, confirming the dataset requires no numerical corrections:

Field	Formula	Mismatches Found
Gross Sales	Units Sold × Sale Price	0
Sales	Gross Sales – Discounts	0
Profit	Sales – COGS	0

2. Profitability by Segment

Segment-level SQL aggregation revealed a sharp divide: four of five segments are healthily profitable, but **Enterprise is the only segment operating at a net loss** (-3.13% margin, -\$614,546 total), despite generating the second-highest sales volume (\$19.6M).

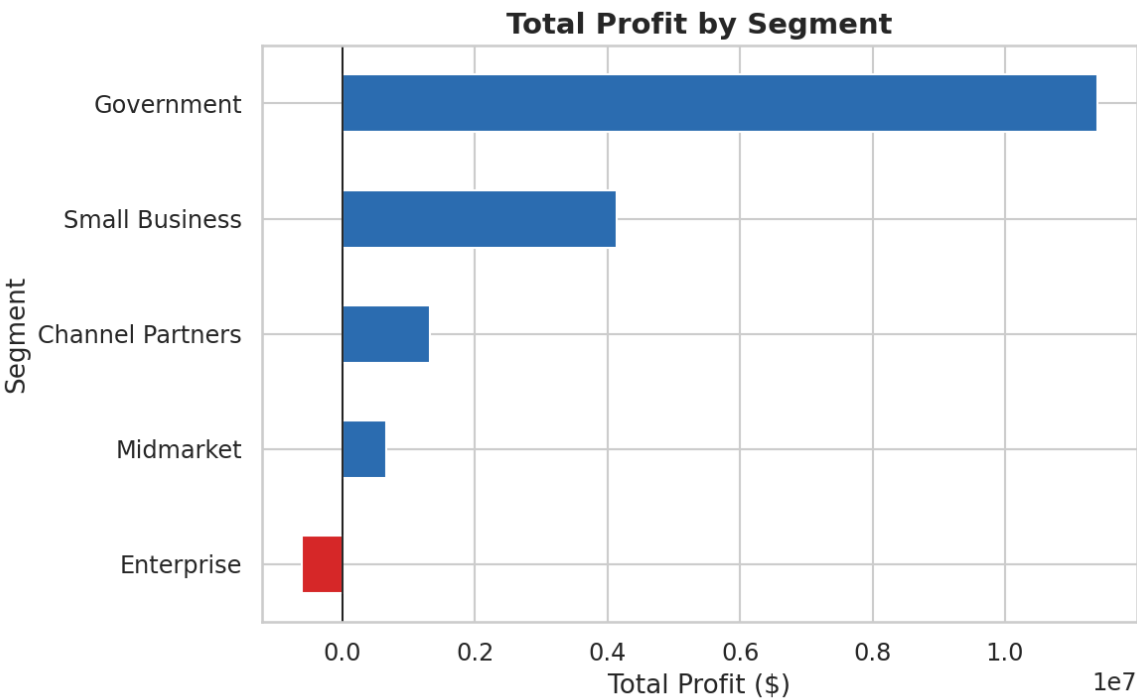


Figure 1: Total profit by segment — Enterprise is the sole segment in the red.

Segment	Sales	Profit	Margin
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Channel Partners	\$1.80M	\$1.32M	73.1%
Midmarket	\$2.38M	\$0.66M	27.7%
Government	\$52.50M	\$11.39M	21.7%
Small Business	\$42.43M	\$4.14M	9.8%
Enterprise	\$19.61M	-\$0.61M	-3.1%

3. Impact of Discounting on Margin

Grouping orders by Discount Band shows a clear, near-linear erosion of margin as discount depth increases — from 21.9% margin on undiscounted orders down to 9.1% on High-band orders (average ~12.5% off).

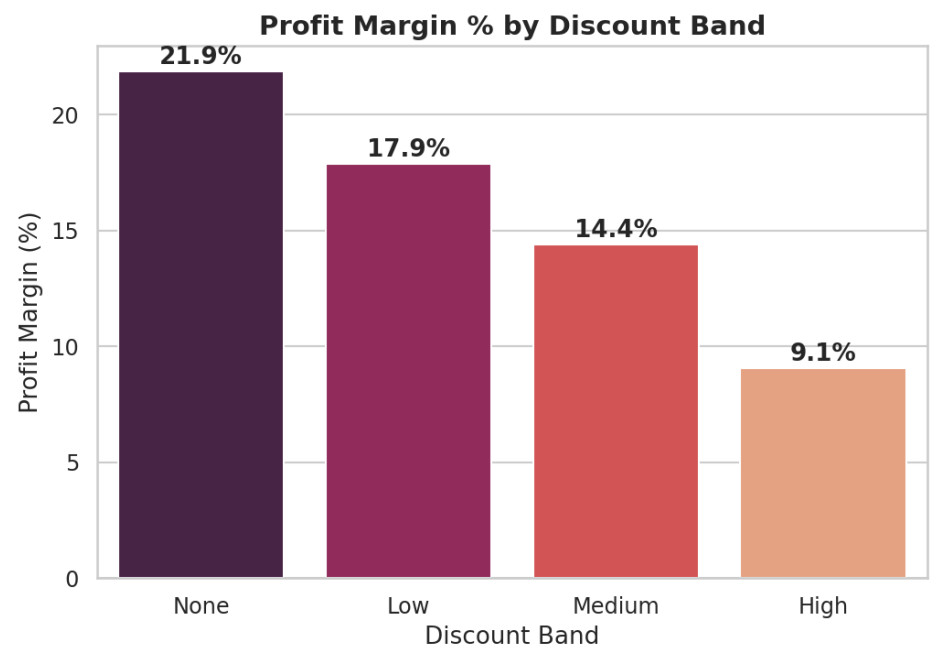


Figure 2: Profit margin percentage falls steadily as discount depth increases.

Cross-referencing segment with discount band pinpoints exactly where the losses originate: Enterprise deals in the Medium and especially High discount bands.

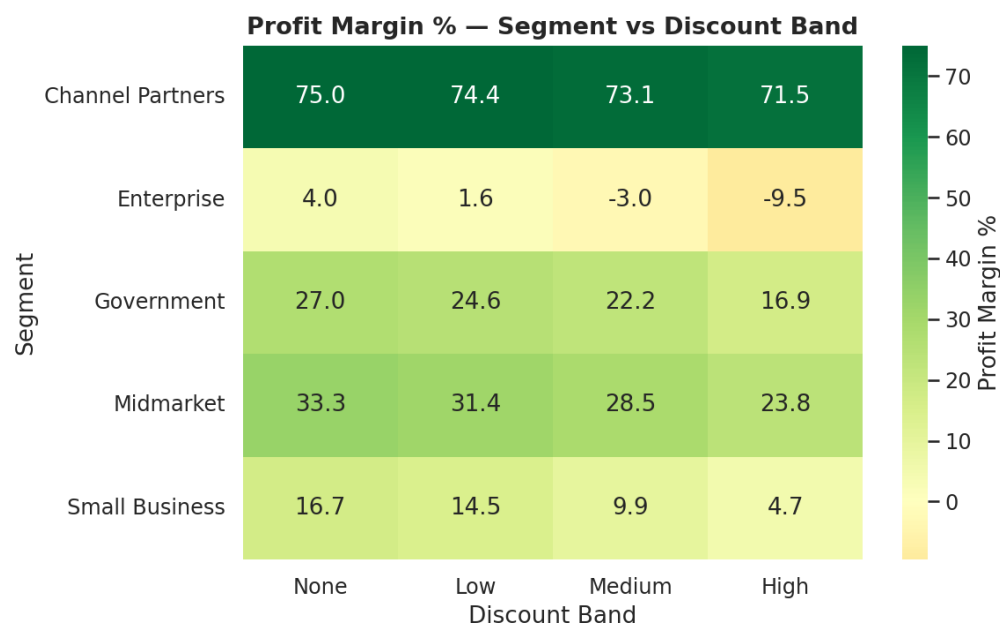


Figure 3: Margin heatmap — red cells (Enterprise × Medium/High) are the only loss zones in the entire dataset.

4. Product and Geographic Performance

Paseo is the clear revenue leader at \$33.0M, more than 60% ahead of the next product, but **Amarilla** converts sales to profit most efficiently at a 15.9% margin. All six products remain profitable in every market, so the loss issue is isolated to Enterprise deal structure rather than product economics.

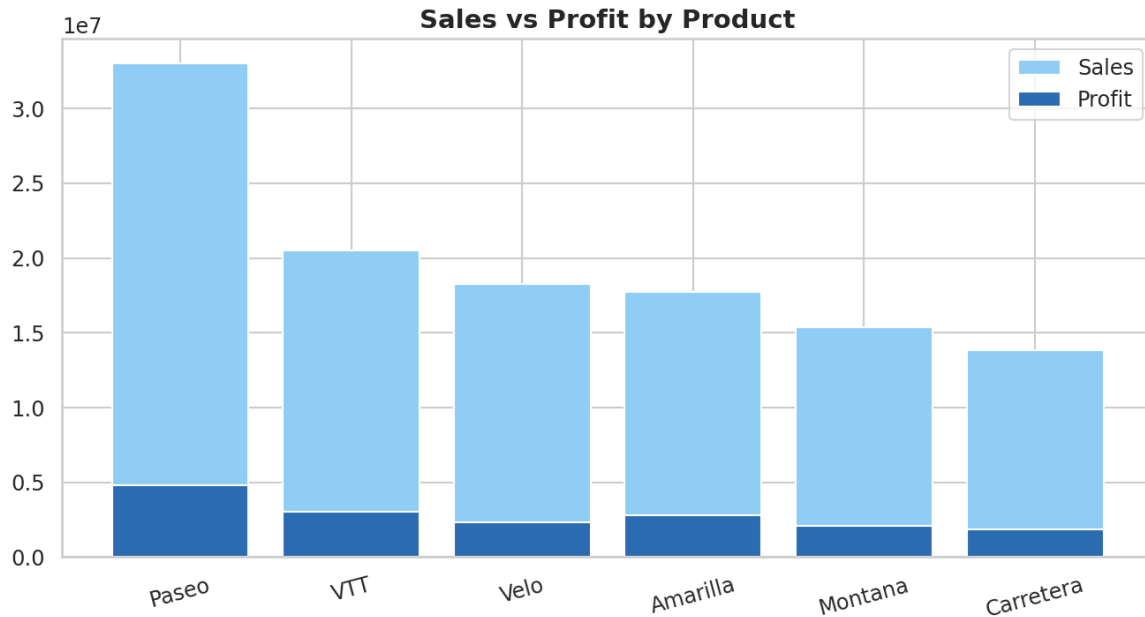


Figure 4: Sales vs. profit by product.

Geographically, all five countries post healthy 12–16% margins with no material outliers — Germany and France edge out the rest at roughly 15.5–15.7%, while the United States has the largest sales base but the thinnest margin (12.0%).

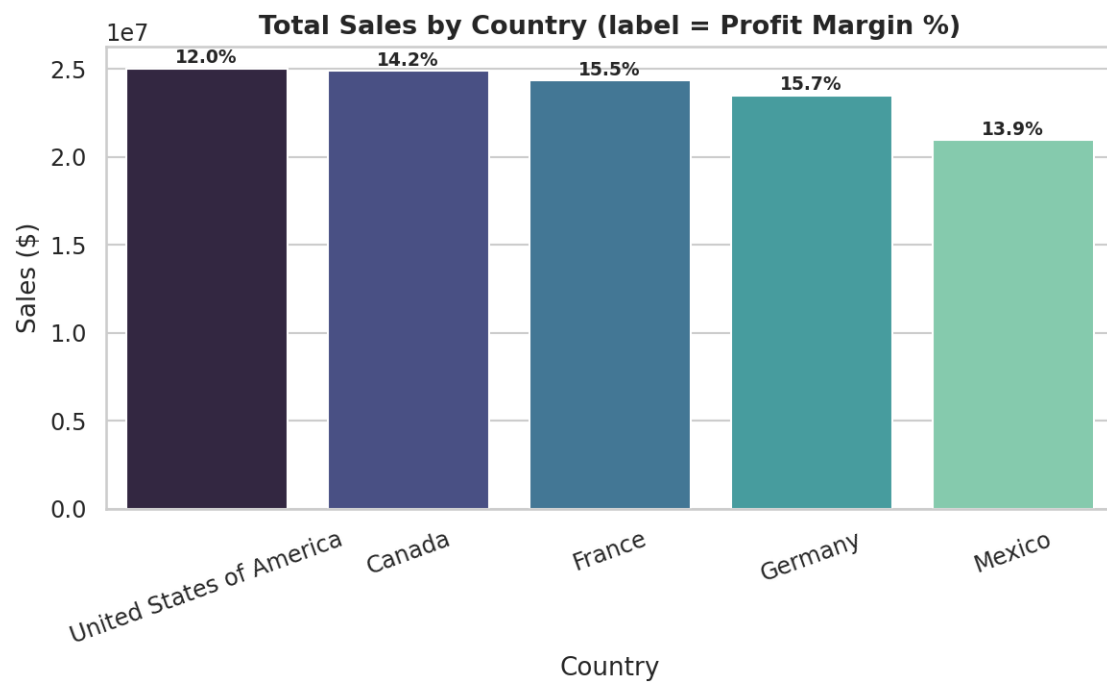


Figure 5: Total sales by country, labeled with profit margin.

5. Sales & Profit Trend Over Time

The dataset spans September 2013 through December 2014. Sales and profit move together month to month, with the strongest performance consistently in October and December of both years — a useful signal for planning inventory, staffing, and campaign timing.

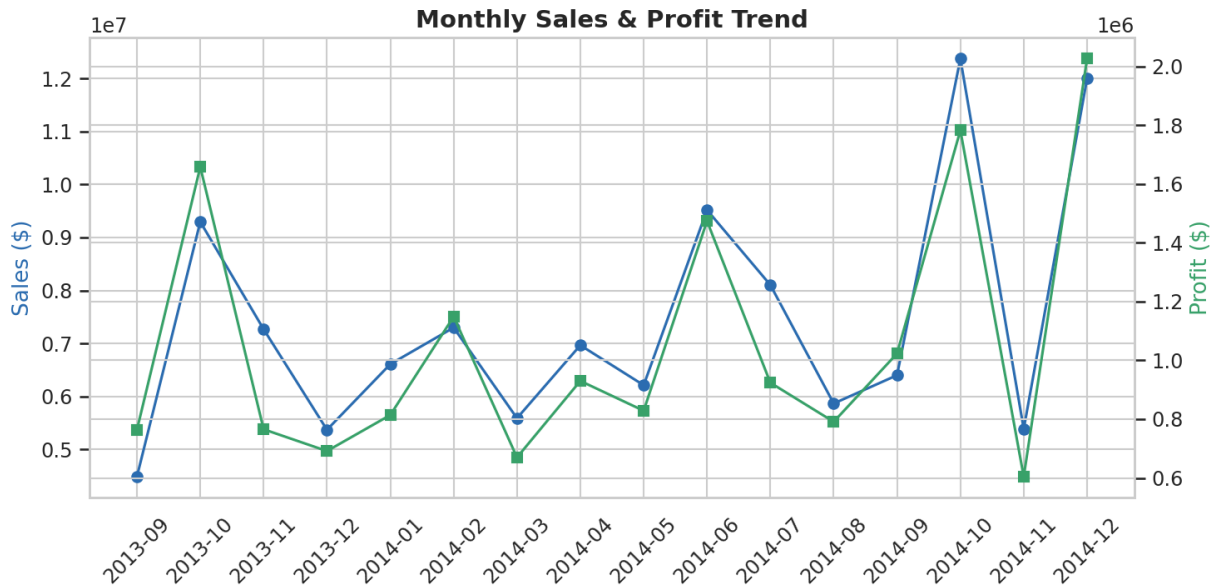


Figure 6: Monthly sales (blue) and profit (green) trend.

6. Key Findings

- Enterprise is the only segment operating at a net loss (-3.13% margin), concentrated in High-discount orders (33 of 58 loss-making orders).
- Channel Partners delivers the best margin in the business (73.1%) despite the smallest revenue base — an efficient, high-value segment.
- Margin declines almost linearly with discount depth: 21.9% (no discount) → 17.9% (Low) → 14.4% (Medium) → 9.1% (High).
- Paseo drives the most revenue (\$33.0M); Amarilla is the most margin-efficient product (15.9%).
- All five countries are solidly profitable (12–16% margin) — geography is not a risk factor.
- October and December are consistently the strongest revenue months across both years covered.

7. Recommendations

- **Cap Enterprise discounting.** Restructure or cap High-band discounts for Enterprise accounts, since this single combination accounts for the business's only losses.
- **Investigate the root cause** of Enterprise's need for deep discounts — likely competitive pricing pressure or long sales cycles — and address it with bundling or value-based offers instead of pure price cuts.
- **Scale the Channel Partners motion.** Its 73% margin suggests a highly efficient go-to-market worth investing in further.
- **Use Amarilla and VTT as margin anchors** in bundles and upsell offers, since they hold profitability best across discount levels.
- **Align seasonal planning with October–December peaks** for inventory, staffing, and marketing spend.

Full code (Jupyter notebook), cleaned dataset, and SQLite database accompany this report as the Complete Code deliverable.